

# SWORD v17c Beta Release Notes

**Version:** v17c beta 0.0.12 **Date:** July 2026 (data exported May 2026) **Base version:** SWORD v17b (March 2025, UNC)

## Changelog

### 0.0.12 (May 2026)

- **Node coordinates restored to v17b for SWOT D0-D2 continuity.** The 344 reaches whose nodes were rederived in 0.0.8 and 0.0.10 now use v17b node `x/y`, `node_length`, `cl_id_min`, and `cl_id_max` values again. A small OC split-revert residue on reach 51111300061 was included in the same pass. Post-revert validation found zero node coordinate differences relative to v17b NetCDF across all 11,112,454 nodes.
- **Centerline-to-node assignments resynchronized.** `centerlines.node_id` was restored from v17b on affected reaches so node geometry and centerline allocation remain internally consistent.
- **Node order repaired on restored-coordinate reaches.** Projection-based ordering over the v17b-restored node coordinates updates `node_order` on 3,725 nodes across 221 of the 344 affected reaches, and updates 45 `dn_node_id/up_node_id` boundary pairs. The repair preserves node coordinates and recomputes the six node distance fields from the current midpoint convention across all 344 restored-coordinate reaches, including a distance-only cleanup for 289 inherited `dist_out` rows on five unchanged-order reaches. AS reach 35301100891 now encodes POM's intended downstream-to-upstream sequence (2, 3, ..., 23, 25, 26, 24, 27, ..., 73, 75, 74, 1) without moving node lat/lon.
- **SWOT slope observations recomputed from node WSE.** `slope_obs_*` on reaches now comes from pass-level regressions of RiverSP node `wse_sm` against the current beta12 `nodes.dist_out`, then reach-level percentiles of those pass slopes. This replaces stale reach-level signed RiverSP slopes whose sign could reflect older node ordering. Median `slope_obs_p50` coverage is 173,732 reaches; negative medians decrease from 18,805 before the correction to 9,988 after the correction. POM's canonical AS reach 35301100891 changes from `slope_obs_p50=-0.0136548684` to `+0.0142687580` m/m while retaining `slope=15.121399` m/km. Downstream users should read both `slope` and `slope_obs_*` directly from 0.0.12; no external sign flip is needed for reaches whose node order changed. The 221 operation-999 `node_order` repair reaches are all in first-digit POM regions 1-6. Test 16 slope-observation and endpoint-WSE diagnostics in first-digit regions 7-9 are separate warning populations, not additional 0.0.12 node-order repairs.
- **Static MERIT/SRTM slope semantics unchanged.** The legacy `reaches/slope` variable remains the nonnegative MERIT/SRTM slope magnitude in m/km; it is not a signed flow-direction variable. 0.0.12 does not change `reaches/slope`, `reaches/wse`, or `nodes/wse` relative to v17b. Endpoint-WSE sign checks may still warn where inherited MERIT/SRTM node WSE values are locally inconsistent with flow direction; those warnings are data-quality diagnostics, not instructions to flip `reaches/slope` or `slope_obs_*`. MERIT/SRTM WSE resampling was evaluated after POM's Test 16 feedback and deferred: the legacy SWORD workflow used UPA-filtered nearest MERIT river pixels and centerline aggregation, so a faithful correction is a science-data reconstruction task rather than a final beta patch.
- **v17c analytic fields preserved.** The revert does not undo the v17c midpoint node-distance convention, lakeflag/type reconciliation, SWOT WSE/width observation statistics, metadata restoration, or export ordering changes. No RiverObs software change or coordinate-substitution mapping table is required for D0-D2 coordinate continuity.
- **Exports regenerated and verified.** NetCDF, GeoPackage, and GeoParquet artifacts were regenerated for all six regions. Export row counts match the DuckDB source for reaches and nodes in GeoPackage/GeoParquet, and for reaches, nodes, and centerlines in NetCDF.

### 0.0.11 (April 2026)

- **Node ordering normalized in NetCDF export.** About 7.5% of reaches (18,553 globally) had nodes stored in descending `dist_out` order in all previous SWORD versions due to arbitrary GRWL centerline digitization direction. Nodes are now exported in ascending `node_order` (downstream-first) for all reaches. This fixes slope sign reversals in processing code that assumes the first node in the array is the downstream end. The underlying data is unchanged; only the file ordering is corrected. This is an export storage-order fix, not a retained topology reversal and not the smaller 0.0.12 operation-999 `node_order` repair set.
- **Restored metadata variables in NetCDF export.** `river_name_local`, `river_name_en`, `version`, `add_flag`, and `swot_obs_source` were accidentally omitted from the export spec in 0.0.4. Restored in 0.0.11 on both reaches and nodes where applicable.

- **Node lakeflag propagated from reach.** Node lakeflag now matches parent reach lakeflag across all 11,112,454 nodes, per JPL request.

#### 0.0.10 (April 2026)

- **Node geolocation fixed on 303 additional reaches.** `rederive_nodes` recomputes node x/y from centerline spatial partitioning for reaches where consecutive node gaps exceed 3x the reach's median spacing and 0.4 km absolute (POM test 6b criteria). By region: AS:189, SA:38, AF:29, EU:21, OC:19, NA:7. Combined with the 41 reaches fixed in 0.0.8, total rederived: 344 reaches (0.14%). **10 reaches were fixed but still trigger the scrambled-node detector** (centerline geometry issues inherited from v17b); see Known Limitations.
- **Reach 35301100891 node\_order rotation and geolocation fixed.** This AS reach had `node_ids` rotated by one position (2, 3, ..., 75, 1 instead of 1, 2, ..., 75) since v17b. Additionally, 74 of 75 nodes were rederived from the centerline (max shift ~12.6 km on node 35301100890011) due to scrambled geolocation detected by POM test 6b. `node_order` now matches `node_id` ascending, `dn_node_id` = 0011, `up_node_id` = 0751. Node `dist_out` and all interpolated distance columns recalculated to match the corrected `node_order`.
- **Reach lakeflag and type reconciled for lake classification.** About 6,200 reaches had inconsistent lakeflag and type fields (lakeflag=1 with type=1, or lakeflag=0 with type=3). Inconsistent reaches were either orphaned (skipped by both river and lake processing) or double-counted. Resolution combined three methods: 1,015 manual reviews through the Streamlit QA app, a gradient-boosted classifier trained on those reviews (82% precision, 6% FPR, applied at high confidence only:  $p > 0.8$  for lake,  $p < 0.2$  for river), and direct corrections from HarP v1.1 lake classifications. Lakeflag and type are now 100% consistent across all 248,673 reaches. The type column now diverges from the reach ID last digit on 2,316 reaches (0.9%) as of 0.0.10 (2,648 by 0.0.12 after subsequent type corrections); the type column is authoritative.
- **Six reaches fixed for N013 closure-bug damage.** Reaches 14278900061 (AF), 31241401301, 48294000081, 45570000125, 34100005185, 42211000503 (AS) had corrupted x/y and `cl_id_min/cl_id_max` from the N013 closure bug (documented in 0.0.8). Nodes rederived from their own centerlines using the fixed code; `node_length` restored from v17b NetCDF to preserve exact sum-equals-reach\_length consistency. Full sweep confirms zero reaches with `node_length` mismatch above 0.1% globally.
- **SWOT reach filters aligned with node filters.** `build_reach_filter_sql` now includes cross-track distance (10-60 km) and valid `time_str` filters, matching the node-level filters. Code change only; no DB data affected.
- **rederive\_scrambled\_nodes.py safeguards added.** After rederiving nodes, the script now automatically recalculates node `dist_out` (prevents N004 violations) and verifies `node_length` sums (catches G002 regressions).

#### 0.0.9 (April 2026)

- **Flow correction fully reverted.** All 810 flow-corrected reaches restored to v17b topology after discovering a scoring tautology in `score_section_confidence` (`slope_from_upstream` = `-slope_from_downstream`). Topology diff vs v17b: 0. `v17c_flow_corrections` table is empty.
- **dist\_out corruption fixed.** 1,976 reaches had BFS-corrupted `dist_out` from a January `CALCULATE_DIST_OUT` bug (wrong outlet, 93–99% error). Restored v17b values. T017 violations: 701 → 557.
- **Pipeline re-run with skip\_flow\_correction=True.** All 6 regions pass all gates.

#### 0.0.8 (April 2026)

- **Node dist\_out unified to midpoint convention.** All six node-level distance columns (`dist_out`, `hydro_dist_out`, `dist_out_dijkstra`, `hydro_dist_hw`, `pathlen_hw`, `pathlen_out`) now use midpoint interpolation from reach-level values. Previously `dist_out` preserved v17b endpoint values while the other five used midpoint, causing a systematic ~100 m discrepancy on single-path networks. On such networks the three outlet-distance columns are now exactly equal at every node. Breaking change from v17b convention; all POM release gate checks pass.
- **Node geolocation fixed on 41 reaches.** `rederive_nodes` recomputes node x/y from centerline spatial partitioning for 41 reaches (SA:5, EU:5, AF:8, AS:20, OC:3) where source NetCDF had contradictory CL-to-node mappings causing >2 km geographic jumps between consecutive nodes. Node lengths now use boundary-to-boundary distances (sum equals `reach_length` exactly). POM example reach 13341000591: max consecutive node jump drops from 13.9 km to 223 m.
- **rederive\_nodes closure bug fixed.** The N013 centerline sync inside `rederive_nodes` captured a stale `reach_id` from an outer loop instead of the current plan's reach. Fixed to use `plan["reach_id"]`.

### 0.0.7 (April 2026)

- **Flow-corrected reach `node_order` and `dist_out` restored.** When pipeline runs skip the facc stage (e.g. when only rerunning distance and mainstem computations), `flipped_reach_ids` was empty, so `update_node_columns` never reversed `node_order` or swapped `dn_node_id/up_node_id` for the 810 flow-corrected reaches. 0.0.6 shipped with `node_order` matching v17b `node_id` order on those reaches (Pierre-Olivier Malaterre detected 389 inverted reaches). 0.0.7 loads the flipped reach IDs from the `v17c_flow_corrections` table when facc is skipped, so the node-order reversal and `dn_node_id/up_node_id` swap always run.
- **Node `dist_out` recomputed for flow-corrected reaches.** After reversing `node_order`, v17b node `dist_out` values are stale on 810 reaches (computed from the wrong flow direction). 0.0.7 recomputes them using cumulative `node_length` midpoint offsets from the corrected downstream boundary, so `dist_out` is monotonic with `node_order` and matches the new flow direction.
- **New lint check N018.** Catches the regression above automatically: flags flow-corrected reaches whose `node_order` is not reversed from v17b `node_id` order. ERROR severity, wired into the POM release gate.

### 0.0.6 (April 2026)

- **Canonical export ordering restored.** The 0.0.5 flat-file exports were written in DuckDB storage order through the generic maintenance exporter, which does not preserve a stable logical row order unless it is told to do so. As a result, node arrays in the distributed files could be split across multiple blocks for the same reach even though the per-row attributes stayed aligned.
- **Nodes now export in reach-contiguous logical order.** 0.0.6 writes nodes ordered by `reach_id`, then `node_order`, then `node_id` across NetCDF, GeoPackage, and Parquet. This restores a clean in-file node sequence for downstream consumers that expect each reach to occupy one contiguous node block and to run from downstream (`node_order=1`) to upstream (`node_order=n`).
- **Observed regression removed.** In Africa, the count of reaches whose NetCDF node rows were split into multiple file blocks dropped from 536 in 0.0.5 to 0 in 0.0.6. Pierre-Olivier Malaterre's example reach 13341000591 now appears as one contiguous node block with `node_order = 1..93`.

### 0.0.5 (April 2026)

- **Promoted corrected bundle to a new version.** The corrected April 2 reissue is now published as 0.0.5 instead of reusing 0.0.4, so testers do not keep hitting stale downloads or cached copies under the same name. This 0.0.5 bundle supersedes all prior 0.0.4 beta uploads on Drive.
- **Release regressions from distributed 0.0.4 artifacts fixed.** The 0.0.5 bundle corrects the specific tester-reported regressions from the prior 0.0.4 uploads: reversed node `dist_out` relative to `node_order`, stale `dn_node_id / up_node_id` orientation on flow-corrected reaches, missing `nodes` layers in some GeoPackages, NULL `dist_out_dijkstra` at isolated ghost coastal outlets, and inconsistent single-node node-distance convention relative to the other node-level outlet distances.
- **Release artifacts renamed and resynced.** NetCDF, GeoPackage, Parquet, release notes, and SHA256 manifests now use 0.0.5 filenames consistently in the local beta folder and Google Drive beta folder.

### 0.0.4 (March 2026)

- **Node propagation to 11.1M nodes.** Five v17c columns (`best_headwater`, `best_outlet`, `pathlen_hw`, `pathlen_out`, `subnetwork_id`) were NULL on all nodes. Now propagated from parent reaches.
- **Dijkstra ghost outlet fix.** Ghost reaches (`type=6`) with `out_degree=0` no longer report `dist_out_dijkstra=0`. All sinks are used as Dijkstra sources for full coverage, with outlet values following the v17b convention (`dist_out_dijkstra = reach_length`). Real outlet counts: NA=7, SA=1, EU=2, AF=1, AS=11, OC=1.
- **Bifurcation routing fix.** At multi-successor nodes, `rch_id_dn_main` now follows the mainstem chain unconditionally (was falling back to score-based ranking, which could disagree with `is_mainstem`). V023 (`pathlen_out` step consistency) violations:  $2 \rightarrow 0$ .
- **hydro\_dist\_hw computed.** Mainstem distance from headwater via `rch_id_up_main` chain walk (mirror of `hydro_dist_out`). Was stale from a prior pipeline run.
- **facc monotonicity fix (T003).** 419 reaches corrected via iterative downstream propagation. T003 violations:  $392 \rightarrow 0$ .
- **Routing weights retrained on effective\_slope.** SWOT `slope_obs_p50` (where reliable and  $n \geq 5$ ) replaces MERIT DEM slope in routing score training. Slope share: 8%  $\rightarrow$  3%, width: 67%  $\rightarrow$  71%. CV accuracy unchanged (88.5%).

- **Node-level distance interpolation.** `hydro_dist_out`, `hydro_dist_hw`, and `dist_out_dijkstra` added to nodes table, interpolated by node position within reach (using v17b `dist_out` offset). `pathlen_hw` and `pathlen_out` changed from flat reach copies to per-node interpolated values.
- **hydro\_dist\_hw convention fix.** Reach-level values shifted from downstream-end reporting to upstream-end reporting, matching `dist_out`, `hydro_dist_out`, and `dist_out_dijkstra`. Headwater reaches now report 0 (was `reach_length`). Node-level values unchanged.
- **F006 junction conservation fix.** 2 remaining junction violations (OC 53130100215, AS 45311901585) resolved by setting downstream `facc` to sum of upstream `facc`. F006 violations:  $2 \rightarrow 0$ .
- **13 AS main\_side reverted to v17b.** Reaches had `main_side` changed from 0 (main) to 1 (side) by an undetermined prior operation; 9 of 13 are linear reaches where side-channel classification is impossible.
- **Distance convention documented.** Variable reference now includes a convention table specifying endpoint reporting, zero-point, and ghost reach behavior for all distance variables, plus node-level interpolation formulas.
- **Variable reference updated.** 7 missing variables added, 8 type mismatches fixed, multi-dimensional array documentation corrected. (The NetCDF export itself uses the v17b `cl_ids [2, N]` format.)
- **Node dist\_out reactive recalc fix.** `CALCULATE_DIST_OUT` operations now correctly recalculate node-level `dist_out` by sorting nodes via `node_order` (geometric position) instead of `node_id`. This fixes an issue where flow-corrected reaches had incorrect node distances after recalculation (e.g., downstream-most node getting downstream reach's `dist_out`).
- **April 2 artifact reissue.** The published April 1 0.0.4 files still had stale orientation metadata on 639 flow-corrected reaches. The reissued NetCDF, GeoPackage, and Parquet bundles recompute `node_order`, `dn_node_id`, and `up_node_id` from node `dist_out`, restore `dist_out_dijkstra = reach_length` for isolated ghost coastal outlets, and align single-node `node.dist_out` with the same midpoint convention used by the other node-level outlet distances.
- **Ghost coastal outlet dist\_out\_dijkstra fix.** Isolated ghost coastal outlets (type=6 sinks with upstream neighbors but no path to real hydrologic outlets) now receive `dist_out_dijkstra = reach_length`, matching v17b `dist_out` behavior. Previously these 703 reaches in NA (similar counts in other regions) had NULL, but they should report the reach-level outlet distance to the ocean outlet point.
- **Exports regenerated (April 1 and 2, 2026).** The initial April 1 bundle captured the code fixes above. The April 2 reissue refreshed NetCDF, GeoPackage, and Parquet artifacts plus SHA256SUMS after repairing published `node_order` / boundary-node orientation metadata, ghost coastal outlet `dist_out_dijkstra`, and single-node node-distance conventions.

### 0.0.3 (March 2026)

- **Routing weights learned from human labels.** Replaced the handcrafted lexicographic 3-tuple (`effective_width`, `log_facc`, `pathlen`) with a weighted scalar score trained on 1,967 human-labeled junction decisions:  $'1.97\log_{1p}(ew) + 0.23\log_{1p}(facc) - 0.23\log_{1p}(slope) + 0.23\log_{1p}(pathlen) - 0.29*stream\_order'$ . Two new signals vs prior releases: `slope` (negative = prefer lower gradient) and `stream_order`. All routing functions use the same score to prevent divergence. (Weights retrained in 0.0.4; see above.)
- **Caroline's reviewer fixes synced.** 127 C001 lakeflag fixes (NA) and 10 C004 type fixes (NA) from the Streamlit reviewer app.
- **NA PostgreSQL geometry fix.** 38,696 NA reaches had NULL geometry in PostgreSQL — v17b geometry overwrite via `dblink` silently failed. Rewrote to read from v17b GPKG files on disk with verification gates.
- Fixed `dn_node_id`, `up_node_id`, and `node_order` for 810 flow-corrected reaches where node ordering was stale (based on v17b `dist_out`). 639 reaches now have `dn_node_id != min(node_id)`, reflecting the inverted flow direction.
- Removed redundant `rch_id_up_1..4` / `rch_id_dn_1..4` vector variables from NetCDF export. Only the `[4, N]` matrices `rch_id_up` and `rch_id_dn` are exported, matching v17b format.
- Fixed `swot_orbits` NetCDF type from string back to `int64` (matching v17b).
- Fixed 15 integer columns widened from `int32` to `int64` in NetCDF export (`n_nodes`, `lakeflag`, `n_rch_up`, `stream_order`, etc.). All shared variable types now match v17b exactly.
- Synced `rch_id_up_1..4` / `rch_id_dn_1..4` DB columns from `reach_topology` table (694 were stale after flow corrections).

### 0.0.2 (March 2026)

- Renamed `is_mainstem_edge` to `is_mainstem`

- Mainstem algorithm refactored: computed per `main_path_id` group (see Section 2.1)
- Fixed `n_rch_up`/`n_rch_down` stale counts at 148 flow-corrected reaches
- Recomputed `facc` at 807 flow-corrected reaches using topological propagation; resolves all F006 junction conservation violations and T003 monotonicity violations at these reaches
- OC reach 51111300061 incomplete split fully reverted to v17b state (434 orphan centerlines, 73 orphan nodes restored)

### 0.0.1 (March 2026)

- Initial v17c beta release
- Includes `node_order`, `dn_node_id`, `up_node_id` (node ordering within reaches)

## 1. Overview

SWORD v17c extends v17b with three additions: computed mainstem topology, SWOT observation statistics, and flow accumulation corrections. No reaches, nodes, or centerlines were added or removed. v17c contains the same 248,673 reaches, 11.1M nodes, and 66.9M centerline points as v17b across all six regions (NA, SA, EU, AF, AS, OC).

Each region is distributed as a single NetCDF4 file (`{region}_sword_v17c_0.0.12.nc`). The group structure matches v17b (centerlines, nodes, reaches), and the `area_fits` and `discharge_models` subgroups under reaches pass through from v17b unchanged. Reach and centerline arrays match v17b canonical row ordering. Node arrays are grouped contiguously by `reach_id` and ordered within each reach by `node_order` (downstream to upstream).

Reach coordinate columns (`x`, `y`, `x_min`, `x_max`, `y_min`, `y_max`) match v17b values across all formats (NetCDF, DuckDB, PostgreSQL). Node coordinate columns (`x`, `y`) also match v17b after the 0.0.12 continuity revert.

All new variables use a fill value of -9999 where no observation or computation produced a value.

For a complete variable catalog, see `v17c_variable_reference.md`.

## 2. New Variables

### 2.1 Mainstem Topology (reaches group)

`is_mainstem` is computed per `main_path_id` group: each group (a mainstem path plus its tributary branches) gets one canonical chain, identified by a greedy walk from the group's shared `best_headwater`. At each junction the algorithm selects the upstream branch with the highest weighted routing score:  $2.02 \cdot \log_{10}(ew) + 0.17 \cdot \log_{10}(facc) - 0.08 \cdot \log_{10}(slope) + 0.35 \cdot \log_{10}(pathlen) + 0.23 \cdot stream\_order$ . These weights were learned from 1,967 human-labeled junction decisions via logistic regression on pairwise log10-difference features, using SWOT slope where reliable (else MERIT DEM). The negative slope weight captures the geomorphic pattern that mainstem channels have lower gradients than tributaries. Mainstem reaches within each group are then assigned `rch_id_up_main` / `rch_id_dn_main` from the chain; non-mainstem reaches use the same weighted score. Ghost reaches (`type=6`) are excluded from mainstem but still participate in routing topology. ~10% of mainstem reaches have no mainstem neighbor at `main_path_id` group boundaries — this is expected by design.

| Variable                       | Type    | Units  | Description                                                                                                                                             |
|--------------------------------|---------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>dist_out_dijkstra</code> | float64 | meters | Dijkstra shortest-path outlet distance reported at the reach upstream endpoint (outlet = <code>reach_length</code> ; values retained for ghost reaches) |
| <code>hydro_dist_out</code>    | float64 | meters | Mainstem outlet distance reported at the reach upstream endpoint via <code>rch_id_dn_main</code> chain (outlet = <code>reach_length</code> )            |
| <code>hydro_dist_hw</code>     | float64 | meters | Mainstem headwater distance reported at the reach upstream endpoint via <code>rch_id_up_main</code> chain (headwater = 0)                               |

| Variable                    | Type    | Units  | Description                                                                         |
|-----------------------------|---------|--------|-------------------------------------------------------------------------------------|
| <code>rch_id_up_main</code> | int64   | —      | Main upstream neighbor <code>reach_id</code> (mainstem-preferred)                   |
| <code>rch_id_dn_main</code> | int64   | —      | Main downstream neighbor <code>reach_id</code> (mainstem-preferred)                 |
| <code>best_headwater</code> | int64   | —      | Routing-score-prioritized headwater <code>reach_id</code> for the network component |
| <code>best_outlet</code>    | int64   | —      | Routing-score-prioritized outlet <code>reach_id</code> for the network component    |
| <code>pathlen_hw</code>     | float64 | meters | Cumulative path length from <b><code>best_headwater</code></b>                      |
| <code>pathlen_out</code>    | float64 | meters | Cumulative path length to <b><code>best_outlet</code></b>                           |
| <code>is_mainstem</code>    | int32   | —      | 1 if reach is on a mainstem path, 0 otherwise                                       |
| <code>main_path_id</code>   | int64   | —      | Unique identifier for each mainstem path group                                      |
| <code>subnetwork_id</code>  | int32   | —      | Connected component ID (Pfaffstetter-offset, globally unique; see Section 4)        |
| <code>dn_node_id</code>     | int64   | —      | Node ID at the downstream end of the reach (lowest <code>dist_out</code> )          |
| <code>up_node_id</code>     | int64   | —      | Node ID at the upstream end of the reach (highest <code>dist_out</code> )           |

Eight of these variables also appear at node level. Six are interpolated by node position within the reach: `dist_out`, `hydro_dist_out`, `hydro_dist_hw`, `dist_out_dijkstra`, `pathlen_hw`, and `pathlen_out`. Three are flat copies from the parent reach: `subnetwork_id`, `best_headwater`, and `best_outlet`. Historical flow-correction topology flips were fully reverted in 0.0.9, so the current 0.0.12 database does not retain a flow-corrected reach set requiring reversed `node_id` order.

All six interpolated distance columns use midpoint offsets:  $\text{offset} = \text{cumsum}(\text{node\_length}) - 0.5 * \text{node\_length}$ . `dist_out`, `hydro_dist_out`, `dist_out_dijkstra`, and `pathlen_hw` use  $\text{reach\_value} - \text{reach\_length} + \text{offset}$ ; `hydro_dist_hw` and `pathlen_out` use  $\text{reach\_value} + \text{reach\_length} - \text{offset}$ . This places each node at the geometric center of its `node_length` segment. On a single-path network (no junctions), node-level `dist_out`, `hydro_dist_out`, and `dist_out_dijkstra` are exactly equal.

`node_order` is a node-level variable (not in the reaches table): 1-based position within a reach, ordered by `dist_out` ascending (1 = downstream end, n = upstream end).

**Distance convention note.** These are endpoint-reported reach scalars, not traversal-origin claims. `dist_out`, `hydro_dist_out`, and `dist_out_dijkstra` are outlet-distance values reported at the reach upstream endpoint and assign `reach_length` at the outlet. `hydro_dist_hw` is also reported at the reach upstream endpoint, but measures distance from `best_headwater` and assigns 0 at the headwater. See the variable reference for the full convention table.

## 2.2 SWOT Observation Statistics

Percentile-based summaries computed from available SWOT observations. All percentile, range, and MAD variables share the units of the underlying measurement.

### Reaches and nodes:

| Variable                                                | Type    | Units  | Description                                         |
|---------------------------------------------------------|---------|--------|-----------------------------------------------------|
| <code>wse_obs_p10</code> – <code>wse_obs_p90</code>     | float64 | meters | WSE percentiles (10th through 90th, in steps of 10) |
| <code>wse_obs_range</code>                              | float64 | meters | WSE observation range (p90 - p10)                   |
| <code>wse_obs_mad</code>                                | float64 | meters | WSE median absolute deviation                       |
| <code>width_obs_p10</code> – <code>width_obs_p90</code> | float64 | meters | Width percentiles                                   |

| Variable        | Type    | Units  | Description                     |
|-----------------|---------|--------|---------------------------------|
| width_obs_range | float64 | meters | Width observation range         |
| width_obs_mad   | float64 | meters | Width median absolute deviation |
| n_obs           | int32   | —      | Total SWOT observation count    |

#### Reaches only:

| Variable                    | Type    | Units | Description                                                       |
|-----------------------------|---------|-------|-------------------------------------------------------------------|
| slope_obs_p10–slope_obs_p90 | float64 | m/m   | Slope percentiles                                                 |
| slope_obs_range             | float64 | m/m   | Slope observation range                                           |
| slope_obs_mad               | float64 | m/m   | Slope median absolute deviation                                   |
| slope_obs_adj               | float64 | m/m   | Adjusted slope                                                    |
| slope_obs_slopeF            | float64 | —     | Slope F-statistic                                                 |
| slope_obs_reliable          | int32   | —     | 0 = unreliable, 1 = reliable                                      |
| slope_obs_quality           | int32   | —     | Integer quality category (0–8; see Section 3)                     |
| slope_obs_n                 | int32   | —     | Number of RiverSP node observations used in pass-level slope fits |
| slope_obs_n_passes          | int32   | —     | Number of SWOT passes used                                        |
| slope_obs_q                 | int32   | —     | Bitfield quality flag (see Section 3)                             |

### 2.3 Flow Accumulation Corrections

A two-stage denoise pipeline corrected flow accumulation (**facc**) values to address three systematic error modes in MERIT Hydro’s D8 (eight-direction flow routing) upstream area: bifurcation cloning, junction inflation, and raster-vector misalignment. In the 0.0.12 database, 95,880 of 248,673 reaches (38.6%) carry corrected values (**facc\_quality** = **denoise\_v3**). Uncorrected reaches retain v17b values. See [facc\\_correction\\_methodology.md](#) for the full algorithm description.

| Variable     | Type    | Group          | Description                                                                                     |
|--------------|---------|----------------|-------------------------------------------------------------------------------------------------|
| facc         | float64 | reaches, nodes | Flow accumulation (km <sup>2</sup> ). Corrected values where applicable; v17b values otherwise. |
| facc_quality | int32   | reaches, nodes | 1 = corrected by denoise_v3; fill_value = not flagged                                           |

After correction, junction conservation violations (downstream **facc** < sum of upstream **facc**) are resolved in all regions. (In 0.0.2, **facc** was additionally recomputed at 807 then-flow-corrected reaches via topological propagation; the flow-correction topology itself was fully reverted to v17b in 0.0.9.)

### 2.4 Other New or Updated Variables

| Variable   | Type  | Group   | Description                                                                                                                                                                                     |
|------------|-------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| type       | int32 | reaches | Reach classification (1=river, 3=lake_on_river, 4=dam, 5=unreliable, 6=ghost). Not present in v17b NetCDF; added in v17c so NetCDF users can filter by reach type without needing the database. |
| dl_grod_id | int64 | reaches | DL-GROD (Deep Learning Global River Obstruction Database; He et al. 2025) dam/obstruction ID                                                                                                    |

| Variable               | Type   | Group   | Description                                                                                                                                                                                                                                   |
|------------------------|--------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>edit_flag</code> | string | reaches | Comma-delimited edit provenance tags (e.g., <code>harp_lake,clf_reconcile</code> ). Also contains v17b numeric codes and the literal string <code>NaN</code> (v17b's no-edit placeholder); see the variable reference for the full value list |

### 3. Flag Encoding Reference

#### `facc_quality`

| Value        | Meaning                                                            |
|--------------|--------------------------------------------------------------------|
| 1            | <code>denoise_v3</code> — corrected by the facc denoising pipeline |
| -9999 (fill) | Not flagged; facc unchanged from v17b                              |

CF attributes: `flag_values = [1]`, `flag_meanings = "denoise_v3"`.

#### `slope_obs_quality`

| Value | Meaning                            |
|-------|------------------------------------|
| 0     | reliable                           |
| 1     | <code>small_negative</code>        |
| 2     | <code>moderate_negative</code>     |
| 3     | <code>large_negative</code>        |
| 4     | negative                           |
| 5     | <code>below_ref_uncertainty</code> |
| 6     | <code>high_uncertainty</code>      |
| 7     | <code>noise_high_nobs</code>       |
| 8     | <code>flat_water_noise</code>      |

CF attributes: `flag_values = [0,1,2,3,4,5,6,7,8]`, `flag_meanings = "reliable small_negative moderate_negative large_negative negative below_ref_uncertainty high_uncertainty noise_high_nobs flat_water_noise"`.

#### `slope_obs_reliable`

| Value        | Meaning                                   |
|--------------|-------------------------------------------|
| 0            | Unreliable                                |
| 1            | Reliable                                  |
| -9999 (fill) | Not computed (no SWOT slope observations) |

#### `slope_obs_q` (bitfield)

| Bit | Value | Meaning              |
|-----|-------|----------------------|
| 1   | 1     | Negative slope       |
| 2   | 2     | Low number of passes |
| 3   | 4     | High variance        |
| 4   | 8     | Extreme value        |
| 5   | 16    | Clipped              |

| Bit | Value | Meaning |
|-----|-------|---------|
|-----|-------|---------|

Flags combine by addition. A value of 0 indicates no quality issues. Example: 5 = negative slope (1) + high variance (4).

**is\_mainstem**

| Value | Meaning          |
|-------|------------------|
| 0     | Not on mainstem  |
| 1     | On mainstem path |

#### 4. Known Limitations

- **SWOT observation coverage:** SWOT statistics are fill\_value (-9999) for reaches and nodes lacking SWOT data.
- **facc correction scope:** 95,880 reaches corrected (38.6%); the remaining 152,793 retain v17b values. Node-level facc propagates from the parent reach.
- **Lake sandwich corrections:** 1,252 reaches were reclassified to lakeflag = 1 where a narrow, shorter-than-neighbor reach sat between lake reaches. The later lakeflag/type reconciliation (0.0.10) rewrote many edit\_flag tags, so 483 reaches carry a lake\_sandwich tag in the 0.0.12 database. ~1,755 similar cases remain (narrow connecting channels, chains).
- **HarP lake corrections:** 7,425 reaches were reclassified from lakeflag = 0 (river) to lakeflag = 1 (lake) based on HarP v1.1 (Hydrography and River Planform) lake classification data, with child node lakeflag updated to match (node lakeflag matches the parent reach on all 11,112,454 nodes as of 0.0.11). The later lakeflag/type reconciliation (0.0.10) folded many of these into combined tags, so 3,981 reaches carry a harp\_lake tag in the 0.0.12 database; node edit\_flag is not tagged. Tags are comma-delimited when multiple apply.
- **area\_fits and discharge\_models:** Direct copies from v17b. Not recomputed against v17c facc or SWOT values.
- **subnetwork\_id vs network:** subnetwork\_id uses Pfafstetter- offset enumeration (globally unique). v17b network uses per-region 1-based IDs. Different component counts (v17c finds more via weakly connected components; 19 subnetworks span multiple v17b networks). network is retained unchanged from v17b.
- **Flow correction fully reverted:** experimental flow-direction corrections (810 reaches at peak, including 389 with ambiguous oscillating WSE slope signals) were fully reverted to v17b topology in 0.0.9 after a scoring tautology was found. v17c-0.0.12 topology is identical to v17b; v17c\_flow\_corrections is empty.
- **main\_path\_id consistency:** 3,134 reaches have main\_path\_id values inconsistent with current (best\_headwater, best\_outlet) tuples (V013- V015 lint checks). 80 reaches in NA have best\_headwater pointing to non-headwater reaches. Requires recomputing main\_path\_id from current headwater/outlet assignments.
- **River naming:** 51.2% of reaches are unnamed (NODATA), ranging from 26% (AF) to 69% (OC). 2.6% of mainstem 1:1 links have local name discontinuities (name changes between adjacent reaches with no junction).
- **Width fill values (A003):** ~1,266 reaches have width=0 (unmeasured) or width=-1 (GRWL lake fill). These are v17b fill values, not data errors. Present in both v17b and v17c unchanged. The A003 lint check is downgraded to WARNING for this reason.
- **Node geolocation corrections deferred:** The 0.0.8 and 0.0.10 rederived-node coordinate edits were reverted in 0.0.12 to preserve SWOT D0-D2 time-series continuity. Some node sequence/geolocation anomalies inherited from v17b therefore remain. Treat full geolocation repair as a v18 or separately approved release-envelope change.

## 5. Quality Audits

Validation checks performed on the v17c data:

| Audit                                            | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Geometry</b>                                  | DuckDB geometries (rebuilt from NetCDF) lack endpoint overlap vertices present in v17b (210,533 reaches affected: 173K +1 point, 37K +2 points). <b>reach_length</b> unchanged. Reach coordinate columns ( <b>x</b> , <b>y</b> , <b>x_min</b> , <b>x_max</b> , <b>y_min</b> , <b>y_max</b> ) copied from v17b to ensure consistency across all formats.                                                                                                                                    |
| <b>Node coordinate continuity</b>                | 0.0.12 restores v17b node <b>x/y</b> , <b>node_length</b> , <b>cl_id_min</b> , and <b>cl_id_max</b> for the 344 previously rederived reaches plus OC reach 51111300061 split-revert residue. Global node coordinate diff vs v17b NetCDF: 0.                                                                                                                                                                                                                                                |
| <b>n_nodes / reach_length<br/>path_freq gaps</b> | Internally consistent. Zero N008/G002/G003 violations. v17b had 4,952 connected non-ghost reaches with invalid <b>path_freq</b> (0 or -9999). Resolved in v17c; remaining nodata values are correctly attributed to ghost reaches ( <b>type</b> =6).                                                                                                                                                                                                                                       |
| <b>subnetwork_id</b>                             | 3,027 components across 248,673 reaches verified. Pfafstetter banding correct. Zero cross-region collisions. 19 subnetworks (0.6%) span multiple v17b networks (expected).                                                                                                                                                                                                                                                                                                                 |
| <b>Topology integrity</b>                        | T001 ( <b>dist_out_dijkstra</b> monotonicity), T012 (referential integrity), T013 (self-reference), T014 (bidirectional): all pass. T005/T007: zero non-reciprocal edges (151 from incomplete flow correction revert resolved in beta 0.0.1). Historical flow-correction topology flips were fully reverted in 0.0.9; <b>v17c_flow_corrections</b> is empty in the current database. Node-level <b>dist_out</b> is recomputed for all reaches using the midpoint interpolation convention. |
| <b>n_rch_up/n_rch_down</b>                       | 148 scalar count mismatches corrected (flow corrections flipped <b>reach_topology</b> but did not recalculate counts). Zero mismatches across all 248,673 reaches.                                                                                                                                                                                                                                                                                                                         |
| <b>OC reach split revert</b>                     | Incomplete <b>break_reaches()</b> split of OC reach 51111300061 (434 orphan centerlines, 73 orphan nodes) fully reverted to v17b state.                                                                                                                                                                                                                                                                                                                                                    |
| <b>River name formatting</b>                     | 291 formatting issues corrected (separators, whitespace). Automated checks now enforce “;” separator and alphabetical ordering.                                                                                                                                                                                                                                                                                                                                                            |
| <b>Flow direction</b>                            | Experimental topology flips were ultimately reverted. The 1,112-flip experiment caused ~30K disconnected reaches and was rolled back; the later retained flow-correction family was also fully reverted in 0.0.9 after the scoring tautology was found. Current v17c-0.0.12 does not retain topology that differs from v17b because of this flow-correction pipeline.                                                                                                                      |
| <b>HarP lake corrections</b>                     | 7,425 reaches reclassified lakeflag 0 to 1 from HarP v1.1 data, with node lakeflag propagated from parent reaches. After the 0.0.10 lakeflag/type reconciliation rewrote tags, 3,981 reaches carry a <b>harp_lake</b> tag in 0.0.12.                                                                                                                                                                                                                                                       |
| <b>lakeflag/type consistency</b>                 | Resolved in 0.0.10. All 248,673 reaches now have consistent <b>lakeflag</b> and <b>type</b> ; <b>type</b> is authoritative and diverges from the reach ID last digit on 2,648 reaches (1.1%) in 0.0.12 due to in-place corrections.                                                                                                                                                                                                                                                        |

For POM (Pierre-Olivier Malaterre) validation results, see `pom_validation_report.md`.

## 6. File Formats

v17c beta 0.0.12 is distributed in five formats. The NetCDF files are the canonical release artifact (full group structure including centerlines and v17b subgroups); the other formats carry reaches and nodes with geometry.

- **NetCDF4:** `{region}_sword_v17c_0.0.12.nc`, one file per region, where region is `na`, `sa`, `eu`, `af`, `as`, `oc`
  - Groups: `centerlines`, `nodes`, `reaches` (plus `reaches/area_fits` and `reaches/discharge_models` subgroups from v17b)
  - Ordering: reach and centerline arrays match v17b canonical ordering; node arrays are reach-contiguous and sorted by `node_order`
  - Fill value: -9999 for all numeric variables (int32, int64, float64)
- **GeoPackage:** `sword_{REGION}_v17c_0.0.12.gpkg` per region (reaches and nodes layers)
- **Shapefile:** `{region}_sword_{reaches,nodes}_hb{XX}_v17c_0.0.12.shp`, split by HydroBASINS Pfafstetter level-2 basin within each region (v17b convention; required by the shapefile 2 GB size limit). Shapefile DBF format truncates attribute names to 10 characters, so fields use abbreviated names; the mapping table `shapefile_field_name_mapping.csv` ships alongside the shapefiles. The NetCDF/GeoPackage/GeoParquet names are authoritative.
- **GeoParquet:** `sword_{REGION}_v17c_0.0.12_{reaches,nodes}.parquet` per region
- **DuckDB:** `sword_{REGION}_v17c_0.0.12.duckdb` per region, with `reaches` and `nodes` tables (geometry stored as GEOMETRY type; written with DuckDB 1.3 — open with DuckDB  $\geq 1.3$  and the spatial extension)
- **Checksums:** SHA256 hashes for all distributed files listed in `SHA256SUMS_0.0.12.txt`

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## 7. Methodology Documentation

| Document                                           | Description                                                  |
|----------------------------------------------------|--------------------------------------------------------------|
| <code>facc_correction_methodology.md</code>        | Facc denoise algorithm, detection rules, correction model    |
| <code>pom_requests_summary.md</code>               | POM validation check tracker (19 checks, production results) |
| <code>pom_validation_report.md</code>              | POM validation production results                            |
| <code>v17c_variable_reference.md</code>            | Complete variable catalog for NetCDF export                  |
| <code>SWORD_v17b_Technical_Documentation.md</code> | v17b baseline reference                                      |